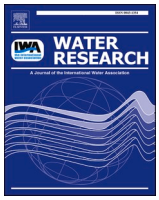
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Strategy to enhance semi-continuous anaerobic digestion of food waste by combined use of calcium peroxide and magnetite

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ABSTRACT

Optimizing methane production from food waste (FW) efficiently is always a hot topic in the field of anaerobic digestion (AD). In this study we aimed to improve the conversion of organics to methane by using CaO_2 and magnetite to enhance the semi-continuous AD of food waste. Under the organic load of 2.5 g VS/L·d^{-1}, the specific methane yield was increased from 333.9 mL CH_4/g·VS to 423.4 mL CH_4/g·VS by adding 0.01 g/L CaO_2 with 0.4 g/L magnetite, improving the production of methane from FW. We assessed reactor performance, ORP changes, mass balance, enzyme activities and characterized the metagenomic profile of microorganisms involved in digestion. These microorganisms showed rapid conversion of volatile fatty acids and increased expression of genes related to hydrolysis and acid production. Thus, the addition of CaO_2 and magnetite optimized the relationship between fermentation bacteria and methanogenic archaea to enhance the overall production of methane. Microorganisms evolved unique adaptive mechanisms in the co-operative environment of CaO_2 and magnetite, as their energy metabolism patterns combined those controlled by individual CaO_2 and magnetite addition. This method of combining a micro-aeration environment with conductive materials provides a new perspective for optimizing the AD of FW.

1. Introduction

In May 2020, atmospheric CO_2 concentrations reached 417 ppm, which is the highest value ever recorded (Negri et al., 2020). It is reported that 6.7% of the global greenhouse gas (GHG) emission from human activities is contributed by food waste (FW) (Pramanik et al., 2019; Slorach et al., 2019). FW is organic matter with high moisture content and must be treated properly to avoid potential environmental pollution. In 2020, 125 million tons (Mt) of food waste was generated in China. Its treatment consumed 30.1 Mt oil of fossil fuels and 16.7 Mt of freshwater and released 37.5 Mt of CO_2 (Liu et al., 2022). Due to the abundance of organics contained in FW, which can be converted to methane as a renewable fuel under anaerobic conditions, anaerobic digestion (AD) has attracted a lot of attention and has become the main choice of treatment. Fermenting bacteria convert the organics in FW to methane and carbon dioxide as the final products, through the processes of hydrolysis, acidification, hydrogen and acetic acid production, and methanogenesis. The conversion efficiency depends on the balance between fermentation and methanogenesis, as well as the interaction among microorganisms.Due to the economic benefits related to the recovery of methane, various methods have been employed to enhance the conversion efficiency of organics to methane, including hydrolysis enhancement, building of DIET (Direct Interspecific Electron Transfer) and bio-augmenttation (Wang et al., 2021b; Ren et al., 2020; Zhao et al., 2022).

In recent years, application of calcium peroxide (CaO_2) in AD was developed to promote the conversion of organics to methane. Calcium peroxide (CaO_2) is a white or yellowish crystalline powder, which is considered to be an environmentally safe inorganic peroxide due to its stability and low cost. It can slowly release $\cdot O_2^-$, H_2O_2, $Ca(OH)_2$ and oxygen at a controlled rate when in contact with water (Zhang et al., 2015), which can be used in pre-hydrolysis of some organic wastes being resistant to digestion. Wang et al. (2019b) pretreated the excess sludge with CaO_2 and found that the CaO_2 promoted WAS decomposition by releasing higher quantities of soluble organic matters for the subsequent digestive process, thus increasing the cumulative methane yield. Tian et al. (2021) used CaO_2 to pretreat wheat straw and enhanced the degradation of cellulose and the release of carbohydrates, resulting in an

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Table 1
Characteristics of the FW and the inoculated sludge.

Characteristics	Feedstock	Inoculum
TS (g/L)	125.70± 2.80	76.00±1.30
VS (g/L)	120.90±3.00	27.40±0.50
VS/TS(%)	97.60	25.60
pH	4.90	7.50
TCOD (mg/L)	162,833 ±4249	37,941 ±2517
SCOD (mg/L)	48,240±1400	1651±327
Crude fat(g/L)	20.03 g ± 0.50	NA
Crude protein(g/L)	18.40±1.70	NA
Carbohydrate(g/L)	70.36±4.00	NA
VFA/ALK (g acetate equivalent/g $CaCO_3$ equivalent)	NA	0.10 ± 0.00

*Data shown as mean ± standard deviation of triplicate samples. NA: Not available.

improved digestive performance. However, the direct use of CaO_2 as an additive for AD has been avoided due to the potential toxicity of the oxidative agents to anaerobic microbes. The suitable ORP of hydrolytic and acidogenic bacterium is below −285 mV, and that of methanogens is below −350 mV (Vongvichiankul et al., 2017; Khanal and Huang, 2003). Therefore, proper application of CaO_2 can adjust ORP in anaerobic digesters, create conditions suitable for facultative microorganisms and promote the growth of hydrolytic and acidogenic bacterium (Wang et al., 2021a; Ping et al., 2020). If the CaO_2 dosage is well controlled, it is possible to improve the digestive efficiency by enhancing the hydrolytic-acidification process to provide a more abundant intermediate for subsequent methanogenesis. However, the related study is still limited, especially in the digester running in a semi-continuous mode.

Due to the collaboration between fermentation bacteria and methanogenic archaea, maintaining high activities of archaea would improve the efficiency of methanogenesis, as would the enhancement of DIET through the use of exogenous conductive materials (Kumar et al., 2021). The presence of conductive materials can build an electron bridge between syntrophic oxidation bacteria (SOB) and hydrogenotrophic methanogens (HMs) accelerates the transformation of electrons from SOB released in fatty acids oxidation to HMs, and avoids the mass transfer resistance through the path of IHT (Interspecific hydrogen transfer). In previous studies, different kinds of conductive materials including activated carbon (AC), graphite, and iron oxides et al. have been adopted (Kumar et al., 2021; Xu et al., 2020). Among them, magnetite poses the most potential due to its excellent conductivity and magnetic separation characteristics. The application of magnetite in promoting AD efficiency has been widely investigated. Wang et al. (2018a) showed that magnetite facilitated the establishment of DIET between fermentative bacteria and methanogens, and it alleviated the need for pili and c-type cytochromes in DIET. As a result, energy metabolism could be effectively used for other cellular activities.

Since CaO_2 addition enhances the growth of fermentation bacteria and magnetite addition is helpful for methanogens' metabolism, will microbes produce different responses if they are added to AD system at the same time? Will the digestive efficiency be further improved under their co-effects? Will magnetite have a positive effect on the regulation of calcium peroxide? Up to now, there is no related report. Therefore, this study tries to employ CaO_2 coupled with magnetite to regulate the digestive system by using FW as feedstock in semi-continuous AD. Digestive performance, carbon mass balance analysis, enzyme activity measurement, and microbial community analysis were used to improve the utilization and metabolism of organic matter. Metagenomic analysis was also used to clarify the internal mechanism of microorganisms in different environments. Results of this study may provide new insights and references for further improving AD efficiency.

2. Materials and methods

2.1. Feedstock and inoculum

FW in this study was prepared in our laboratory to ensure the uniformity of the feedstock mixture in the semi-continuous digester. It was composed of 400 g/kg rice, 300 g/kg cabbage, 200 g/kg pork and 100 g/kg tofu. These raw materials were obtained from a vegetable market in Chongqing, China. The crude materials were pulverized after being cooked and then mixed with ultrapure water to make 2 L modular food waste.Inoculated sludge was obtained from an anaerobic digester in Tangjiatuo Wastewater Treatment Plant in Chongqing. Before using the inoculum, it was placed at 37°C for at least 7 days, so that the residual organic matter can be exhausted. CaO_2 and micron magnetite were purchased from Aladdin, Shanghai. Characteristics of the FW and the inoculated sludge are shown in Table 1.

2.2. Experimental setup

Continuous stirred tank reactor (CSTR) with a total volume of 2 L was built for this study. The working volume of the reactor was 1.8 L with a headspace of 0.2 L. It was equipped with a hydrothermal heating jacket to keep the temperature at 35±1 °C, and an ORP monitoring sensor to record ORP variation during digestion. The digester was operated under a semi-continuous feeding mode to maintain a constant hydraulic retention time (HRT) of 20 days with the organic load rate (OLR) of 2.5 g VS/L·d^{-1}. Subsequently, 0.01 g/L CaO_2 and 0.4 g/L magnetite were added into the reactor (R3) to examine their potential promotion effect. Reactors for comparison were named as R1 with 0.01 g/L CaO_2 addition and R2 with 0.4 g/L magnetite. CaO_2 or magnetite was not added to the control group, and the rest of the conditions were kept the same as in the experimental groups. The dosage is referred to the previous study with slight adjustment (Baek et al., 2017; Tian et al., 2021). Samples were taken daily, then adding FW, magnetite and CaO_2 to the original concentration. To ensure a completely anaerobic environment, nitrogen was charged into the reactor for 5 min per day after feeding. Each experiment was repeated three times.

2.3. Analysis methods

2.3.1. Chemical and physical analysis

Conventional parameters (TS, VS, TAN, TCOD, SCOD) were determined by the standard method developed by APHA 1995. pH value was measured by pH meter (Leici, PHS-3S, Shanghai). On-line ORP measurement instrument (Sinomeasure, SIN-ORP-6050-5 M, Hangzhou) was used. Biogas in digestion was collected by a 4 L air bag and its specific volume was measured daily through drawing the gas off by a graduated syringe. Biogas composition, as well as volatile fatty acids (VFAs) in slurry, was analyzed by a gas chromatograph (Jiedao GC1690, Hangzhou) equipped with a thermal conductivity detector (TCD) and flame ionization detector (FID), related analysis method can be found in our previous study (Zou et al., 2018).

Relative activities of α-glucosidase, protease, acetate kinase, Coenzyme F420, and cytochrome C were determined by Elisa kit (Hengyuan biological, Shanghai). The bacterial electron transport system (ETS) activity was determined by the reduction of 2-(p-iodophenyl)-3-(p-nitrophenyl) -5-phenyltetrazole chloride (INT, an exogenous electron acceptor) to DIRTY (INF) (Wan et al., 2016). Carbon content in the slurry was determined by an elemental analyzer (Vario EL III, Elementar, Germany), and the carbon mass balance was built based on references (Nguyen et al., 2019).

To measure the free radicals (i.e. ·OH and $\cdot O_2^-$) released by CaO_2 decomposition in the digestive mixture, EPR (Electron paramagnetic resonance) tests were performed (Bruker EMXplus, Germany) and the detailed operations are described in Supporting Information.

To evaluate the effect of different additives on the survival efficiency

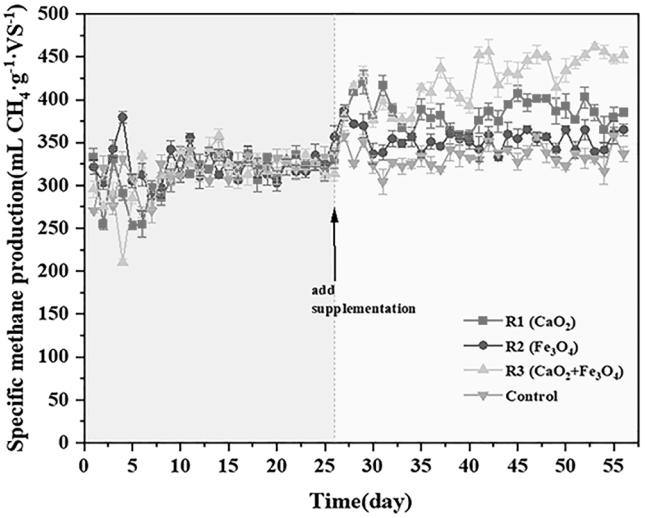


Fig. 1. Daily specific methane yield for different digesters.

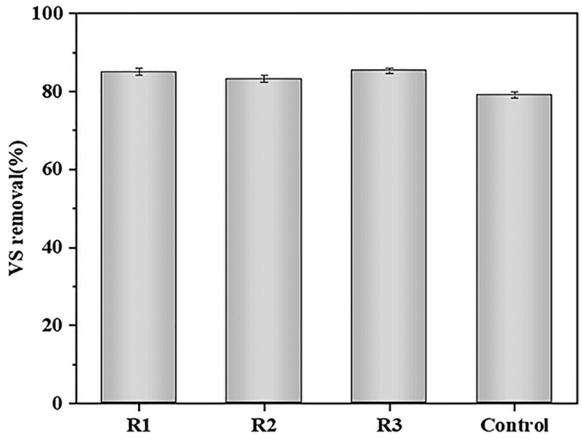


Fig. 2. VS removal in different digesters.

of microorganisms, flow cytometry (BD, AccuriTMC6 Plus, USA) was applied to measure the integrity and damage percentage of microbial cells. Specific operations referred to previous studies (Du et al., 2021).

2.4. Microbial community and metagenomic analyses

Sludge samples were collected from the stable period ($p<0.05$ for inter-group comparison and $p>0.05$ for intra-group comparison) for microbial analysis. All samples were refrigerated at −20 °C for subsequent DNA extraction. Genomic DNA was extracted using a DNA extraction Kit (E.Z.N.A.® Soil DNA Kit, Omega bio-tek, America) using Majorbio's Illumina MiSieq platform. The V3-V4 hypervariable region of 16S rRNA gene was amplified by PCR using bacterial primers 338F (5′ -ACTCCTACGGGAGCAGCC-3′) and 806R (5′ -GGactachVGGGTWTC-TAat-3 '). The v3-V5 region of 16S rRNA gene was amplified by Arch344F (5 '-ACGGggyGCAGCAGGCGCGA-3 ') and Arch915R (5′ -GTGCTCCCCCGCCAATTCT-3 '). The amplified products were purified and quantified, and pyrosequencing was performed on Illumina Miseq PE300. Analysis of microbial community structure and diversity on the free online Majorbio Cloud Platform (www.majorbio.com). For metagenomic analysis, all procedures are given in the Supporting Information.

2.5. Data analysis

All physicochemical and enzymatic data are presented as the mean ±standard deviation. To investigate whether there are significant differences in the performance of different reactors, an analysis of variance (ANOVA) was employed to evaluate the significance of results. The test method is *student'st*-test, and the confidence interval is 95%. If the p value is less than 0.05, the corresponding results are considered statistically significant. In this study, IBM SPSS 27.0 statistic was used for analysis and calculation. Origin pro was used to draw graphics and error bars.

3. Results and discussion

3.1. Digestive performance

3.1.1. Methane production

Biogas is critical to the AD process and directly reflect digestive performance. The four reactors were operated continuously for 56 days and the daily methane yield is presented in Fig. 1. During the first 26 days, the four groups ran in parallel, maintaining the same OLR of 2.5 g VS/L·d^{-1} with the HRT of 20 d There were large fluctuations in daily methane yield during the first 7 days' of digestion, after which the yield got stabilized of all experimental groups on day26 ($p>0.05$), different additives were introduced and digestive performances were evaluated. As seen in Fig. 1, the addition of magnetite(R2) slightly improved (5.95%) the methane yield, with an average daily specific methane production of 353.6 mL CH_4/g VS. Jung et al. (2020) also found that the promotion of methane production by the addition of single magnetite was also not significant. Compared to these slight improvements, the addition of CaO_2(R1) significantly increased the daily methane yield to 384.9 mL CH_4/g VS ($P>0.05$), with was 15.3% greater than the control digester. Notbaly, the combination of CaO_2 and magnetite(R3) resulted in the highest methane production, with an average yield of 423.4 mL CH_4/g VS, which is 26.8% higher than the control.

To further explore the positive effects of these additives on methane production, we recorded the cumulative production of methane during the first 12 h of digestion (Fig. S1) was recorded. Supplementation with CaO_2 and/or magnetite improved the methane production rate in the following order: R3>R1>R2>Control. We also measured the total volatile fatty acid (TVFA) change (Fig. S2), which is linked to methane production. During the first 12 h, all four reactors showed an initial increase in TVFA concentrations, but the time at which the peaks appeared and the peak concentration varied. R2 produced the highest TVFA concentrations, peaking at 6 h of digestion, verifying that the addition of magnetite can promote VFA production, consistent with previous literature reports (Peng et al., 2018). However, the consumption rate of the TVFA in this digester is quite limited. The peak concentration in digester R3 appeared 1–2 h after their addition, and values were lower than for R1 and R2, suggesting there was a faster conversion of the intermediates into the methane. After 12 h of digestion, the TVFA concentrations decreased to a low level, revealing that most of the organics contained in feedstock were degraded.

3.1.2. Organics degradation

Fig. 2 shows the average removal rate of VS in different digesters. It can be seen from the figure that the participation of these additives improves the utilization of VS. The digesters in which CaO_2 was involved (R1 and R3) showed higher removal rates of VS. The VS removal rates were 84.9%, 83.2%, 85.3% and 79.1% for R1, R2, R3 and the control group, respectively. Thus, all combinations of additives improved the degradation rate of organic matter, and the greatest improvement was seen with the combined use of CaO_2 and magnetite.

3.1.3. ORP variations and free radicals release

ORP is defined as the tendency of a solution in acquiring or losing electrons. Solutions with a positive ORP have stronger oxidation conditions, while a negative ORP is indicative of a reduction environment.

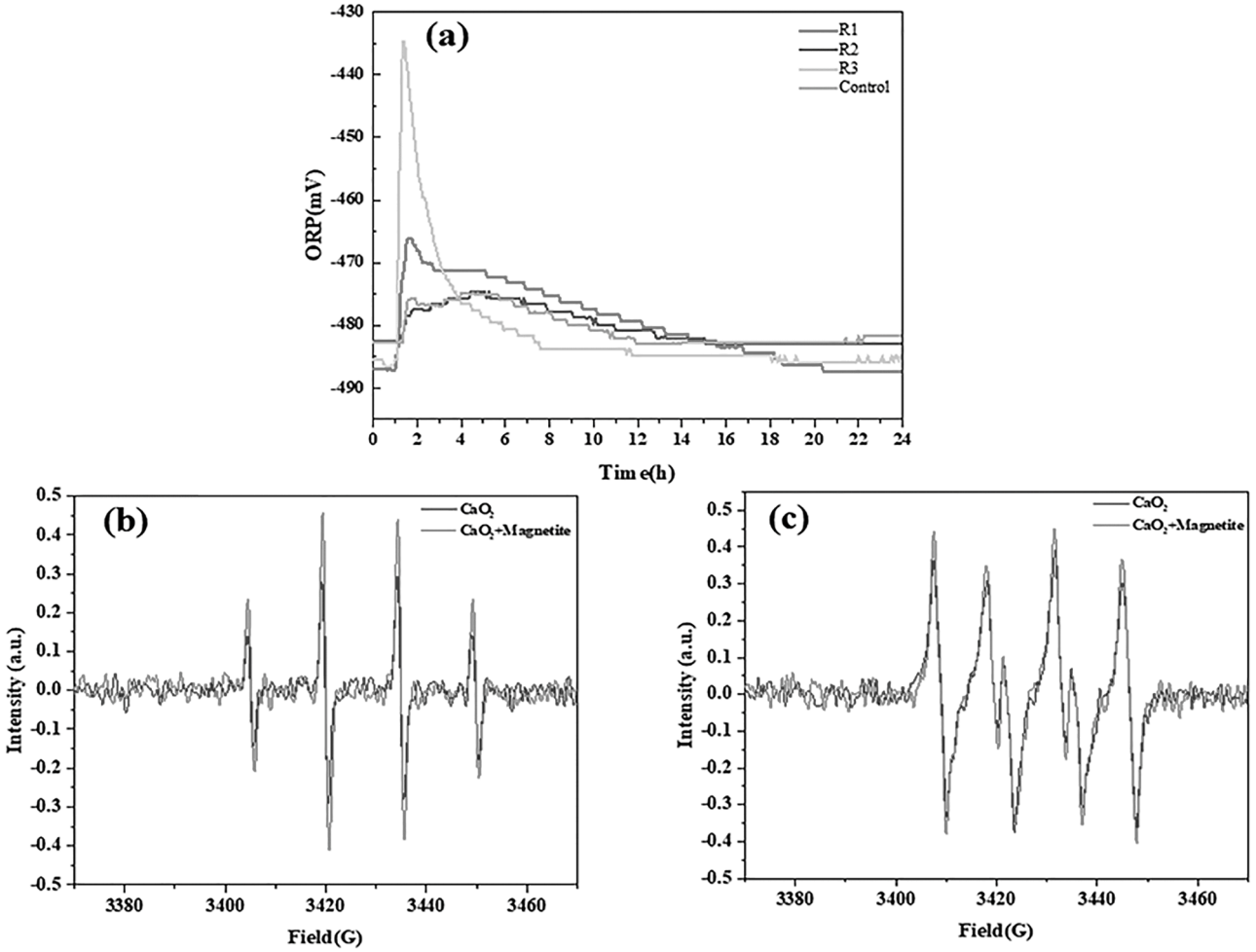


Fig. 3. ORP variation in 24-hour period after feeding (a); relative strength of ·OH (b) and ·O2− (c).

Due to the participation of CaO_2, which can be used as an oxidant, ORP in digesters would suffer fluctuation, and this fluctuation was recorded and presented in Fig. 3. The presence of CaO_2 caused a rapid increase in the ORP, reaching peak values after 2 h, followed by a gradual decline. Use of CaO_2 alone (R1) resulted in a peak ORP value of −466 mV, which was about 20 mV increase from the ORP baseline. When magnetite and CaO_2 were used simultaneously (R3), the ORP increased more significantly from −485 mV to about −435 mV and dropped rapidly within 2 h after reaching the peak value. These results indicate that there might by a catalytic effect of the magnetite in decomposing CaO_2, which accelerated the release of hydrogen peroxide, as well as some oxidative free radicals such as ·OH and $O_2\cdot^-$ and reduced the retaining time of the promoted ORP. Compared to CaO_2 addition alone, this kind of fast ORP stimulation may be beneficial to the hydrolysis of organics and the enriching of different functional microorganisms, which will be discussed in Section 3.5.

To verify the catalytic effect of magnetite on CaO_2 decomposition, EPR paramagnetic resonance detection was carried out. The relative intensities of ·OH and $\cdot O_2^-$ in digester R1 and R3 were measured (Fig. 3b and c). The results show that the vibration of ·OH in R3 is stronger than R1, while there was also a slight enhancement of the $\cdot O_2^-$ vibration in R3. These results verify that the combined use of CaO_2 and magnetite will produce more ·OH in a shorter time, creating a fast ORP increase-drop environment.

3.2. Carbon mass balance

During AD, it is desired that as much carbon contained in the feedstock can be converted to C-CH_4 by microorganisms as possible. Therefore, to determine the effect of adding trace amounts of CaO_2 or magnetite and their combination on the carbon conversion, a carbon mass balance analysis was built, and the results can be found in Fig. 4. Compared with the control digester, the individual addition of CaO_2 and magnetite increased the carbon recovery in the form of CH_4, which

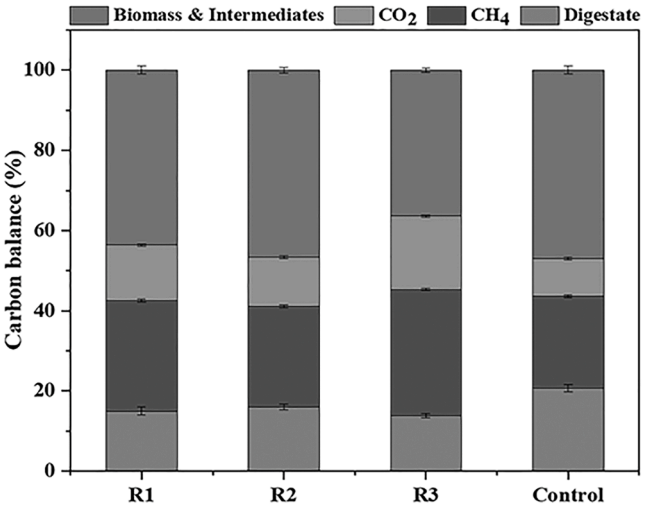


Fig. 4. Carbon mass balance in different digesters.

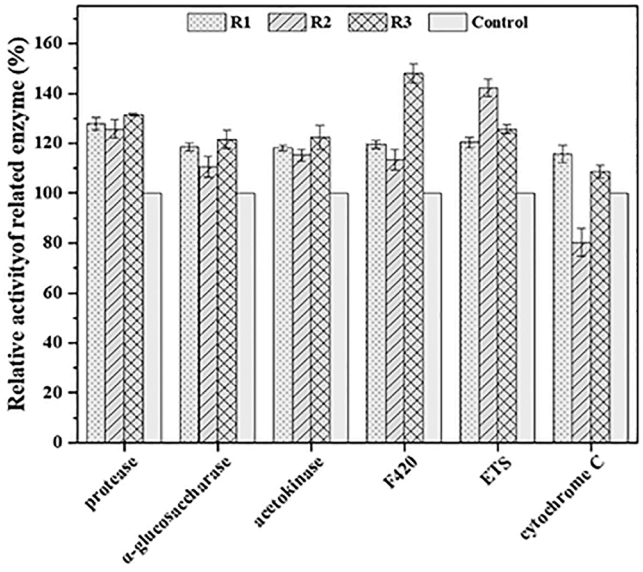


Fig. 5. Relative activities of key enzymes and ETS based on the control digester.

consisted of 27.6% and 25.2% in total. These findings were also supported by improved VS removal (Fig. 2). The simultaneous use of the CaO_2 and magnetite further improve the conversion of organics to the CH_4, and the percentage of the C-CH_4 reached 31.5%. For the carbon contained in digestate, the contents in digester R1, R2 and R3 were 15.0%, 16.0% and 13.8%, which was significantly lower than that of the control digester (20.7%). It is known that feedstock organics in the solid form were hydrolyzed into a dissolved form that can be utilized by microorganisms. So, the lower content of C-digestate in the presence of additives verifies that the utilization of substrates by microorganisms was enhanced. In the CaO_2-containing digesters (R1 and R3), less carbon mass was contained as biomass and intermediates. Especially for digester R3, a much lower content (36.4%) was observed. The results imply that the addition of CaO_2 and magnetite together enhances the degradation and utilization of substrates by microorganisms that produce CH_4 and CO_2, and this effect was most significant in the presence of CaO_2. Moreover, synergism exists between CaO_2 and magnetite in the bio-conversion of these organics.

3.3. Enzymes activity and ETS

To compare the metabolic activity of microorganisms in each step when adding CaO_2 or magnetite and their combination, the enzyme activity and ETS in different digesters were measured (Fig. 5). Firstly, hydrolysis-related enzymes including protease and α-glucosaccharase were examined. Previous studies have shown that magnetite accelerated substrate hydrolysis and increased the hydrolase activity (Wang et al., 2018a). In the presence of magnetite alone (R2), the activities of protease and α-glucosaccharase were 125% and 110% of the control digester, respectively. Further enhancement of enzyme activity was seen in the presence of CaO_2 (R1 and R3). α-glucosaccharase and protease can accelerate the transformation of solid macromolecular organic matter into small molecular organic matter for utilization by microorganisms (Wang et al., 2022b). Acetokinase is the key enzyme in acetic acid production and controls the final production of acetic acid (Estevez-Canales et al., 2015). The activity of acetate kinase was improved by these additives, particularly in the CaO_2-containing digesters, indicating a positive effect of electron acceptor in acetate conversion. Coenzyme F420 is commonly used to reflect the methanogenic activity (Mu et al., 2011), and the combination of CaO_2 and magnetite (R3) greatly improved its activity, being about 149% of the control digester. The F420 activity in R1 and R2 was 120% and 113% of the control. The order of the coenzyme F420 activity was highly consistent with the results presented in Section 3.1.1.

ETS is an indicator of metabolic activity and is related to electron transfer between cells (Du et al., 2021). The presence of magnetite significantly enhanced the ETS of the digester, suggesting that magnetite improves cellular respiration. The lower ETS values and higher methane production efficiency in digesters R1 and R3 can be ascribed to the presence of different ecological networks in these digesters. Compared to the control group, the ETS in digester R1 with individual CaO_2 addition also got a promotion. O_2 acts as an electron acceptor in ETS (Wang et al., 2020b). The reduction of O_2 in NADH can release a large amount of energy, which then generates a proton gradient. This process can drive the synthesis of ATP and the transport of other metabolites, which may be the reason for the improved ETS by CaO_2. Moreover, more than just facilitating the electron transfer activity, the combination of CaO_2 and magnetite can induce more symbiotic modes among microorganisms.

C-type cytochrome (c-Cyts) in the outer membrane of bacteria directly and indirectly participates in the process of interspecific electron transfer (Estevez-Canales et al., 2015). From the figure, it can be seen that CaO_2 addition increased the expression of c-Cyts. Previous studies have proved that oxygen-containing functional groups contained on the bioelectrode surface enhanced the expression of c-Cyts in biofilm and thus promoted the extracellular electron transfer (Li et al., 2019). It was also reported that appropriate oxygen addition increased the expression of c-Cyts in yeast (Burke et al., 1997). CaO_2 provided a micro-aeration environment, which can reasonably explain this result. However, the addition of magnetite decreased the expression of c-Cyts. This is because magnetite can replace the function of c-Cyts in electron transfer, and thus reducing the energy demand in c-Cyts synthesis (Wang et al., 2019a). c-Cyts content in R3 was between R1 and R2 and was higher than that of the control. This indicates that CaO_2 and magnetite play a co-regulatory role on microbial energy metabolism. Magnetite does reduce the energy required for c-Cyts synthesis by microorganisms to some extent, but CaO_2 seems to play a more dominant role as an energy regulator. Although enhanced ETS implies enhanced efficiency of intercellular electron transfer, no correlation between ETS and C-Cyts has been demonstrated in previous literatures. That is to say, the microorganisms prefer the energy metabolism mode under micro-aerophilic environment in this process rather than the DIET mode mediated by magnetite, which is also confirmed in Section 3.5.2.

Table 2
Biodiversity statistics of 16SrRNA gene library of the bacterial community in different groups.

	Parameters	Sequence number	Shannon	Simpson	ACE	Chaol	Coverage
Bacteria	Control	25679	3.800	0.057	421.044	419.217	0.9975
	R1	29678	3.539	0.076	418.222	418.884	0.9978
	R2	26873	3.731	0.055	420.821	411.714	0.9974
	R3	28415	3.633	0.064	421.722	419.565	0.9975
Archaea	Control	52609	2.321	0.155	68.892	68.143	0.9999
	R1	54593	2.075	0.277	69.513	70.250	0.9998
	R2	52309	1.758	0.390	68.488	67.200	0.9998
	R3	46990	2.085	0.259	69.624	66.875	0.9998

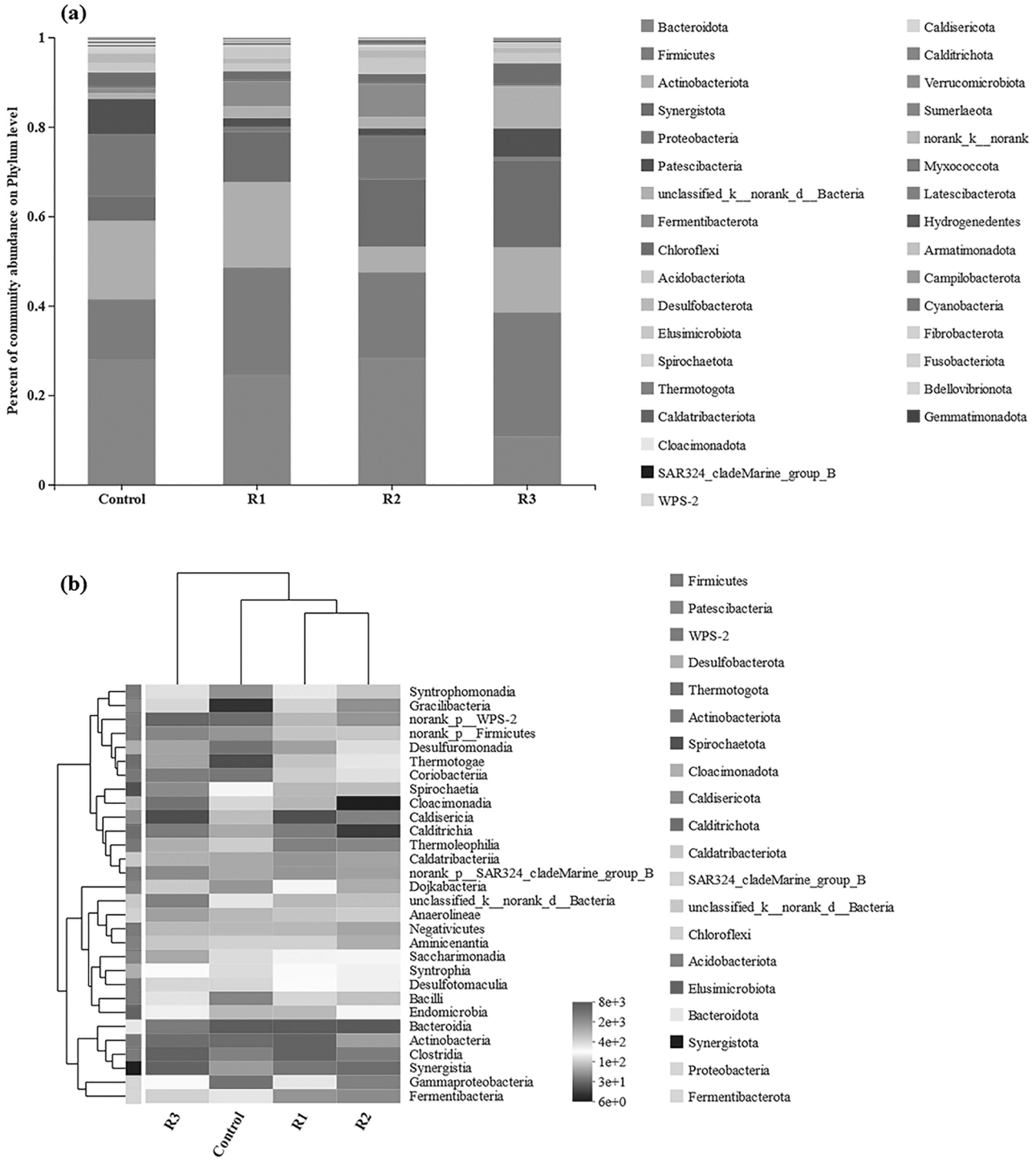


Fig. 6. Taxonomic distribution of bacteria at the phylum (a) and class (b) level.

3.4. Microbial community analysis

Microorganisms are key participants in the decomposition and utilization of organics in AD and their abundance and composition of the microorganism community dictates digestive performance. Thus, microbial community analysis of bacteria and archaea carried out. Indexes of Chao, ACE, Simpson, Shannon and Coverage were used to judge the abundance and diversity of microorganisms in different digesters (Table 2). The index of coverage in every digester was as high as 99%, indicating that almost all of OTUs were detected in this study. According to the Shannon and Simpson index, the individual addition of magnetite had a slight effect on the diversity of bacteria, but it poses a great effect on the archaea community. Addition of CaO_2 alone or combination with magnetite reduced the diversity of microbial bacteria and archaea community, indicating that the microbial bacteria population may evolve in a specific direction.

3.4.1. Bacterial community

The taxonomic distribution of bacteria at the phylum and class level are shown in Fig. 6. *Bacteroidota, Firmicutes, Actinobacteriota, Synergistota* and *Proteobacteria* were the dominant phylum. Other studies have shown that *Bacteroidota* and *Firmicutes* are the main bacteria phylum

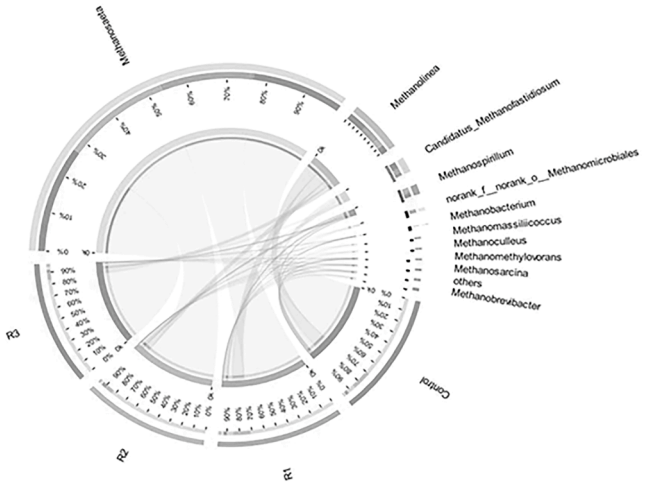


Fig. 7. Relative abundance of archaea at the genus level.

ubiquitous in food waste AD system (Gao et al., 2021; Wang et al., 2020a). *Firmicutes* produces extracellular enzymes (such as cellulase, lipase and protease), which mainly play a role in the metabolism of cellulose, protein, lignin and lipids, and are highly related to the hydrolysis of substrates (Zhao et al., 2020). In R1 and R3 with CaO_2 addition, *Firmicutes* was enriched to a higher degree, reaching 24.2% and 28.0% respectively, and the enrichment degree in R2 also increased to 19.3%. *Firmicutes* is also a kind of facultative bacteria. The micro-aeration environment provided by CaO_2 changes flora towards the direction of higher oxygen tolerance. Therefore, fermentation under micro-aeration conditions can metabolize complex substrates, which leads to a higher hydrolysis rate. Previous studies have also found this phenomenon (Duarte et al., 2018). *Bacteroidota* is responsible for the hydrolysis of carbohydrates to acetate, butyrate and propionic acid (Luo et al., 2015). Its relative abundance in R1-R3 and control was 24.3%, 28.2%, 10. 6% and 27.7%, indicating that the combination of CaO_2 and magnetite altered the abundance of this bacteria the most. *Synergistota*, also named *Synergistetes*, can produce short-chain fatty acids and act as syntrophic acetate oxidizers (Rivière et al., 2009). It occupies a high level in R3 with an abundance of 19.5%. This high abundance would likely help to maintain a low hydrogen partial pressure in the digester, consistent with the results of the higher methane production rate and the relatively lower VFA concentration within 12 h in R3. *Actinobacteriota* can also easily decompose polysaccharides and protein substrates under anaerobic conditions (Zou et al., 2018). In R1 and R3 with CaO_2 addition, this phylum maintained a high abundance. *Probacteoria* decreased significantly in R1 and R3, implying it is quite sensitive to ORP variation brought by CaO_2 decomposition. The abundance of *Elusimicrobiota* in R1 is much higher than that of R2 and R3. This phylum contains a single facultative species, which has a cluster of genes for resisting oxygen stress (Guo et al., 2021). From above results, it can be concluded that the combination of CaO_2 and magnetite promoted both hydrolysis and VFAs syntrophic oxidation.

3.4.2. Archaea community

The relative abundance of archaea at the genus level is shown in Fig. 7. The main archaea in all digesters was *Methanosaeta*, followed by *Methanolinea*, accounting for 87%~92% of the total archaea. *Methanosaeta* is a common archaea species that has a high affinity for acetic acid at low acetate concentration, while *Methanolinea* is a hydrogenotrophic methanogen that uses H_2 to reduce CO_2 to produce methane (Xu et al., 2018; Zhao et al., 2016). The abundance of *Methanosaeta* in the control and R1 was 72.0% and 73.2%, respectively, which was exceeded by the magnetite participated digesters: 85.2%(R2) and 86.1% (R3). In addition, *Methanolinea* accounted for 16.1% and 14.2% of the total archaea population in the control and R1 digester, respectively, while its abundance in magnetite-containing digesters was 6.9% (R2) and 6.1% (R3). It can be inferred from above results that the addition of CaO_2 alone does not significantly change the archaea composition, while the participation of magnetite does. However, the combination of CaO_2 and magnetite did not affect archaea abundance. The enrichment of acidophilic acid methanogen and reduction of hydrogenotrophic methanogen indicate that there is a reduced dependency of the hydrogenotrophic methanogen when magnetite was present. The magnetite in the digestive system built a direct connection between syntrophic oxidation bacteria and hydrogenotrophic methanogens in electron transference, which reduced the yield of hydrogen for hydrogenotrophic methanogens utilization. In addition, the DIET further promoted the syntrophic oxidation of fatty acids and more acetic acid was released, which can be found in Fig. S1 (R2). Zhao et al. (2020) also found that when DIET was induced, the energy metabolism of *Methanosaeta* was more active. In conclusion, *Methanosaeta* abundance increased and *Methanolinea* abundance decreased. Moreover, though the digester R2 (magnetite) and R3 (CaO_2+magnetite) displayed a similar distribution of archaea, they showed different methane and VFAs production profiles, indicating that the methanogenic activities in these digesters differ, which will be discussed in the following section.

3.5. Metagenomics analysis

3.5.1. CaO_2/magnetite improves the level of metabolism and utilization of organic matter

To further clarify the mechanism by which bacteria and archaea enhance hydrolysis and methane production when composed to CaO_2 and magnetite, metagenome sequencing of these microorganisms was carried out. Fig. 8a shows the abundance of key KEGG pathways related to the AD process under different conditions. The presence of these additives increased the abundance expression of most genes, while the abundance promotion follows the order of R3>R1>R2>Control, which was consistent with the methane yield presented in Fig. 1. However, the digester R2 presented higher methane metabolism abundance than R1. From this result, we can infer that CaO_2 addition alone does not significantly regulate methane metabolism. It could however indirectly improve methane production by enhancing both the processes of hydrolysis and acidification. As observed in Fig. 8a, the metabolism of carbohydrates (e.g. starch, sucrose, glucose and pyruvate) and VFAs (e. g. butanoate and propanoate) are promoted. Genes related to VFAs production in fatty acid biosynthesis pathway were further analyzed (Table S1). The abundance of *fabG*-encoding gene was significantly increased in the presence of CaO_2, while the addition of magnetite alone showed a slight increase. It is reported that *fabG* is related to the coding of acyl reductase, which uses NAD+ or NADP+ as the electron acceptor in lipid metabolism and fatty acid synthesis (Zhang et al., 2017). Moreover, the combination of CaO_2 and magnetite also increased the abundance of *fabZ* which is necessary for encoding hydrolase, providing more small molecule substrates for fatty acid biosynthesis. The gene abundance of *bcca* and *accd6* was relatively low, but they still play an important role, because they are functional genes encoding acetyl-CoA carboxylase and they participate in producing VFAs (Ping et al., 2020). Its abundance in digester R1 and R3 was significantly higher than that in the control. *Fas* encodes fatty acid synthetase and participates in organic metabolism. Similarly, CaO_2 has a stronger ability to promote this gene than magnetite. *DesA3* is related to acyl reductase (Kachroo et al., 2006), which acts on paired donors and binds or reduces molecular oxygen directly participating in the formation of fatty acids. In the micro-oxygen environment created by CaO_2 addition, the expression of this gene was more than twice that of the control. These results verify the promotion of fatty acid synthesis in the presence of CaO_2.

CaO_2 also increased the expression of genes related to the citric acid cycle pathway more significantly than individual use of magnetite. Citric acid cycle, also known as TCA cycle, is one of the central pathways

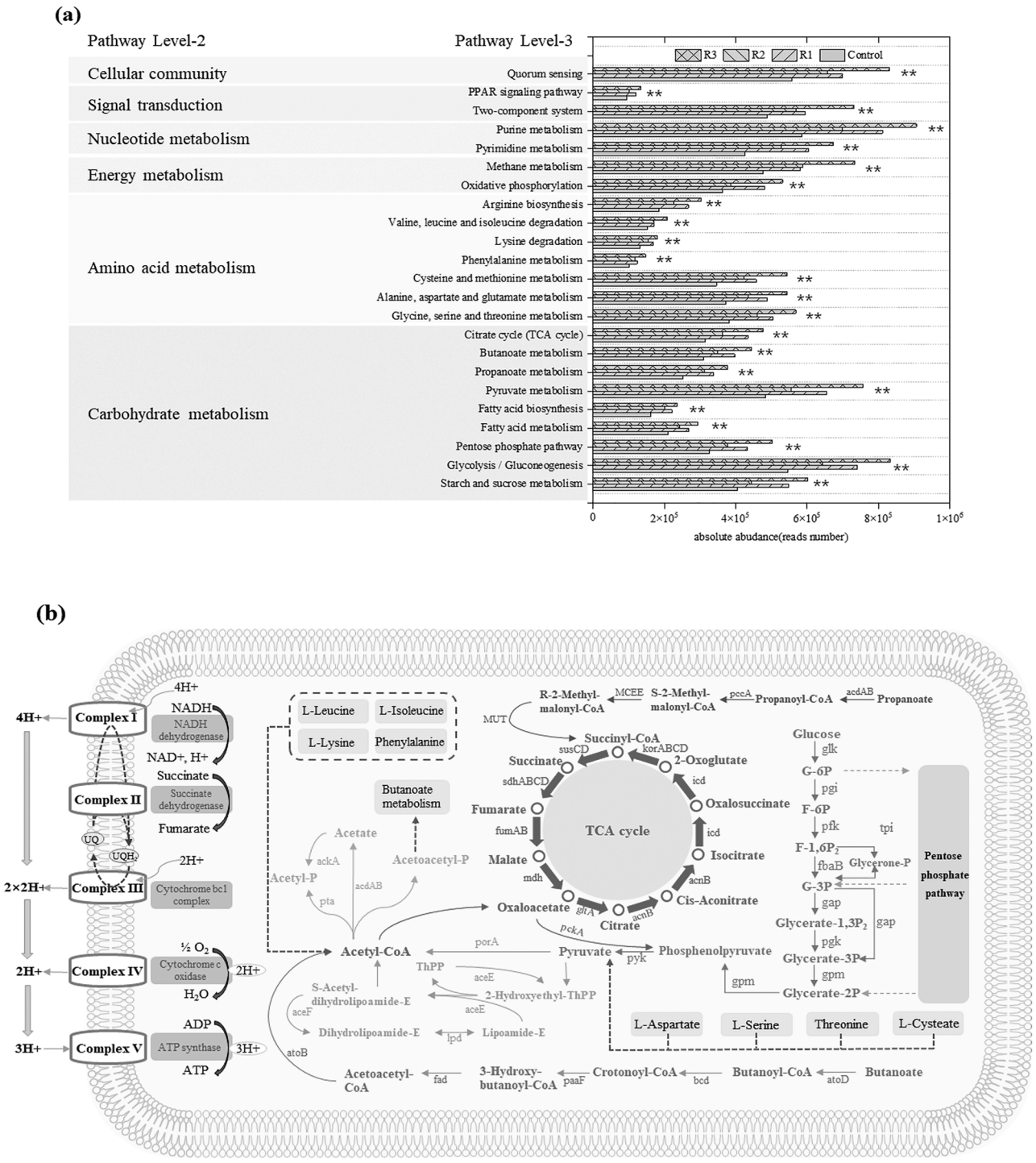


Fig. 8. Effects of CaO_2 and magnetite addition on differential expression of KEGG pathway (a) and organic matter and energy metabolism path map (b). ** indicates significant difference (corrected p values are less than 0.01) between samples from two different conditions.

of metabolism, and is responsible for oxidizing respiratory substances to generate adenosine triphosphate (ATP) in the presence of oxygen (Sweetlove et al., 2010). Fig. 8b shows the TCA cycle pathway and table S7 lists the key enzymes and their expression abundance in the TCA cycle. The enhancement of TCA cycle genes suggested that CaO_2 promotes the aerobic oxidation of carbohydrates. Furthermore, the combination of CaO_2 and magnetite greatly enhanced the expression abundance of this pathway. It can be seen from the Table S2 that CaO_2 also strongly enhances the expression of TCA cycle-related modules, and that is further enhanced by the combination of CaO_2 and magnetite. The top 5 modules associated with the TCA cycle were Gluconeogenesis (M00003), Pyruvate oxidation (M00307), Citrate cycle (M00009), the first carbon oxidation (M00010) and the second carbon oxidation (M00011). Acetyl-CoA is the substrate for the TCA cycle and an essential intermediate product for methane production. Sugar, fat and protein all need to enter the TCA cycle and undergo oxidative phosphorylation through acetyl-CoA, which is completely oxidized to generate CO_2 and H_2O, and release energy for ATP synthesis. Acetyl-CoA usually comes from three pathways, including the degradation of ketogenic amino acids, β-oxidation of fatty acids and glycolysis products (Sweetlove et al.,

2010). The addition of magnetite could promote the formation of pyruvate (M00001) by promoting glycolysis, and the expression of all related enzymes in this metabolic process was significantly elevated (Table S7). Pyruvate is the main precursor for acetyl-CoA production, implying that there are more materials in R2 that can be utilized by microorganisms and converted to methane and CO_2. At the same time, we also found that the ability of magnetite to regulate the conversion of pyruvate to acetyl-CoA (M00307) by microorganisms was difficult to improve, and the expression of most enzymes in this process was similar to that of the control. Therefore, the organic matter metabolic process in R2 was slower compared to R1 and R3. In digester R1 and R3 with CaO_2 addition, pyruvate metabolism was strongly increased. As the final product of glycolytic pathway, pyruvic acid can be oxidized into acetyl-CoA in mitochondria, and then finally to CO_2 and H_2O to complete the aerobic oxidation and energy supply process of glucose. Pyruvate can also realize the mutual transformation of sugar, fat and amino acids in vivo through acetyl-CoA and TCA cycle. In other words, although magnetite can regulate the process of microbial metabolism of glucose to pyruvate, but in the process of entering the TCA cycle microbial energy supply is insufficient, so the problem of metabolic retardation occurs. In contrast, the metabolism of organic matter was enhanced, as well as the regulation of metabolic synthesis and adaptability to the micro-oxygen environment. Especially, when CaO_2 and magnetite were co-regulated, the stimulation of metabolism of microorganisms may be more intense, and more organic matter is rapidly converted to pyruvate and synthesized into acetyl-CoA, thus increasing methane production.

As an important component of FW, protein can be transformed by hydrolysis and for subsequent amino acid metabolism, another major metabolic pathway of microorganisms. Amino acids can be converted into pyruvic acid by deamination, transamination and decarboxylation. Then VFA is produced through the metabolism of pyruvate. In addition, some amino acids can also be converted into acetyl-CoA, which is an essential intermediate product for subsequent methane production (Wang et al., 2022a). Therefore, the conversion of amino acids to pyruvate or acetyl-CoA is highly relevant. Pyruvate or acetyl-CoA are involved in 10 amino acid metabolic pathways, among which there are 6 pathways with complete conversion from amino acids to intermediate products (pyruvate or acetyl-CoA). The amino acid metabolic pathways in Fig. 8a include six pathways, which are Glycine, serine and threonine metabolism (ko00260), Alanine, aspartate and glutamate metabolism (ko00250), Phenylalanine metabolism (ko00360), Lysine degradation (ko00310), Valine, leucine and isoleucine degradation (ko00280) and Cysteine and methionine metabolism(ko00270). The expression of these metabolic pathways in CaO_2 participated digesters (R1 and R3) is higher than that in R2 with magnetite alone. According to KEGG database, apart from converting amino acids into acetyl-CoA, these pathways are also involved in pyruvate metabolism, fatty acid degradation, glycolysis and other processes. Under the action of CaO_2, the gene expression of these important amino acid metabolic pathways was significantly improved, which is indicative of more intermediate metabolites (eg. Acetyl-CoA) for subsequent methane production in the AD process.

3.5.2. CaO_2/magnetite triggers higher energy metabolism

Pentose phosphate pathway (PPP) is an important process of glucose −6- phosphate (G6P) metabolism to produce NADPH and ribose 5-phosphate (ribose 5P). Therefore, this pathway can provide reducing power for various cell synthesis reactions (such as oxidative phosphorylation) (Rauch et al., 2010). At the same time, PPP can compensate and coordinate glycolysis and TCA cycle, supporting the regulation of the microbial community (Seiboth and Metz, 2011). The expression abundance of genes related to the oxidative phosphate pathway is listed in Table S3. The data indicate that the combination of CaO_2 and magnetite may improve the energy metabolism of microorganisms by driving the expression of oxidative phosphate pathways.

In cells, ATP can be transformed with ADP to storage and release energy, thus ensuring the energy supply for all life activities of cells, and its synthesis depends on the transmembrane gradient of Na^+/H^+ (Zhao et al., 2020). Generally, bacteria mainly express genes encoding F-type ATPase to catalyze ADP to ATP (M00157), while archaea mainly express genes encoding V/A-type ATPase to catalyze ADP to ATP (M00159) (Bird et al., 2011). Magnetite and CaO_2 promoted the gene module encoding F-type ATPase, but there was no significant difference in their gene abundance, while the gene encoding V/A-type ATPase shows higher abundance in the presence of CaO_2. Therefore, although CaO_2 can hardly change the community structure of archaea (Fig. 7), the energy metabolism of archaea becomes more active in the micro-oxygen environment than in the environment only assisted by magnetite, and their combination further increases this effect.

Oxidative phosphorylation was greatly promoted in the CaO_2-containing digesters (Fig. 8a), but was limited in the presence of magnetite alone. It has been reported that the amount of energy produced by substrate horizontal phosphorylation in the absence of oxygen is rather limited, with only four ATP molecules per molecule of glucose. However, in aerobic conditions, oxygen, as the terminal electron acceptor, can produce 28 additional ATP molecules in each molecule of glucose through glycolysis, TCA cycle and oxidative phosphorylation. That is to say, the micro-oxygen environment introduced by CaO_2 stimulates the fermentation bacteria and the acidogenic bacterium to produce higher energy metabolism. The key enzymes in the oxidative phosphorylation pathway include NADH dehydrogenase, succinate dehydrogenase, cytochrome bd complex and cytochrome C oxidase, which act as front-end complexes and provide power for ATP synthase to convert ADP into ATP. Individual CaO_2 addition increased the expression of all these key enzyme synthesis genes. For the NADH dehydrogenase, its gene abundance of R1 is significantly increased. As the produced NADH tends to be reduced to NAD+, the feedback inhibition caused by the accumulation of NADH/NAD+ ratio can be avoided, thus promoting the related carbohydrate metabolism(Lin et al., 2016). The transformation between succinic acid and fumaric acid cannot directly cause the transmembrane gradient of Na^+/H^+ to produce ATP and transfer electrons but can drive the transformation between ubiquinone (UQ) and UQH_2, and transfer $Na^+/H^+$and electrons from NADH dehydrogenase complex to cytochrome bd complex (Zhao et al., 2020). Succinate dehydrogenase is an important membrane-bound enzyme in TCA cycle. The increase of its expression may be related to the promotion of TCA cycle by CaO_2, which may accelerate the transport of $Na^+/H^+$and electrons in mitochondrial inner membrane, thus further accelerating ATP synthesis. Cytochrome C oxidase can reduce O_2 to H_2O, and its increased expression in R1 means that the micro-oxygen environment may stimulate higher energy metabolism. Interestingly, addition of magnetite only increased the gene abundance of ATP synthase gene, while the gene expression of the four key enzymes decreased compared with the control group. Magnetite, as an excellent conductive material, can directly conduct electrons from fermentation bacteria to methanogenic archaea, reducing energy consumption for electron transfer components such as cytochrome C and e-pili (Wang et al., 2018b). This also explains the decrease in abundance of the cytochrome bd complex and cytochrome C oxidase. Furthermore, it seems that magnetite can also replace part of the functions of NADH dehydrogenase and succinate dehydrogenase, and directly act as the carrier of electron transfer, further reducing energy loss. More interestingly, the combination of magnetite and CaO_2 seems to make microbes more "smart". Although magnetite also exists in R3, microorganisms seem to prefer the way of energy synthesis and metabolism assisted by CaO_2, and retain some ways of saving energy expenditure of magnetite, because the expression of genes encoding cytochrome bd complex, succinate dehydrogenase and cytochrome C oxidase presented significant synergistic effect of individual CaO_2 and magnetite.

Moreover, additives also increased the abundance of genes responsible for pyrimidine and purine metabolism, which are related to cell activity (Wang et al., 2018a). Especially, in CaO_2 participated digester,

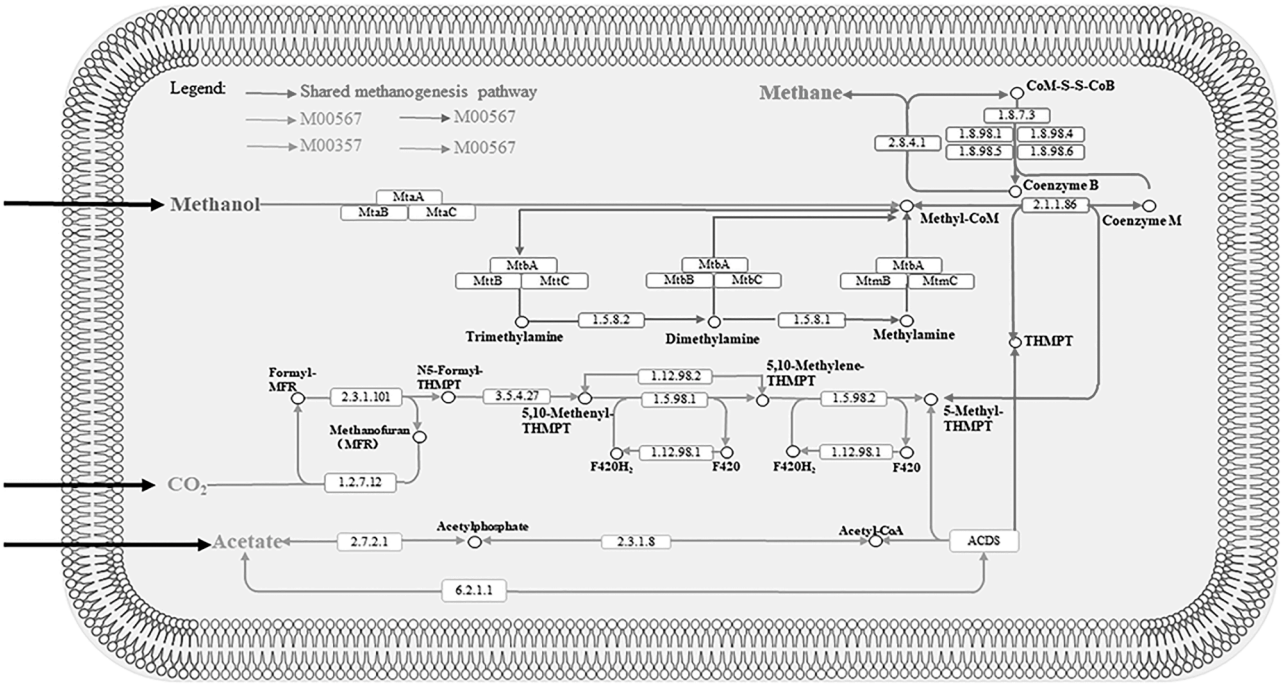


Fig. 9. Methane metabolic pathway map.

the abundance of *msmX* was greatly improved, especially when CaO_2 is combined with magnetite, which is twice of the control digester (Table S1). *MsmX* is responsible for encoding ATP-binding protein of polysaccharide transport system, which has been shown to combine ATP hydrolysis with the active transport of ions, sugars, amino acids, lipids and other substrates (Wilkens, 2015). From above results, it can be concluded that the combination of CaO_2 and magnetite not only improves the energy metabolism of microorganisms, but also accelerates the transformation of available substances.

3.5.3. CaO_2/magnetite stimulates the strong expression of methanogenic metabolism modules

Further analysis of the metagenomic sequencing results was conducted to determine the impact of CaO_2 and magnetite on methanogenic pathways. In the metabolic pathway of methane, archaea can use four methane generation modules: acetic acid (M00357), CO_2 (M00567), methylamine/dimethylamine/trimethylamine (M00563) and methanol (M00356). Fig. 9 shows the specific pathway of methane metabolism. Table S4 shows the abundance of these four modules, while Table S5 shows the abundance of related enzymes specifically involved in metabolic pathway. Interestingly, the addition of CaO_2 alone increased the expression of the M00357 module, while the expression of the other three modules decreased. The presence of magnetite clearly increased the expression of these four modules, which was un line with the reported positive effect of magnetite on archaea. Additionally, there is no obvious difference between the reads of R1 and control. Although it is less than R2, the gene abundance the methane metabolic pathway increased in the R1 digester (Fig. 8a), further indicating that CaO_2 does not regulate methanogenic archaea. The increase in gene abundance of this metabolic pathway may be caused by the enhancement of the metabolism of bacteria that provides better interaction with methanogenic archaea. It appears that the symbiotic network of bacteria and archaea moves in different directions under the micro-oxygen condition introduced by CaO_2. In addition, it seems that the combination of CaO_2 and magnetite compensates for the deficiency caused by CaO_2 addition alone, further strengthening the expression of genes related to methane-producing modules.

Carbon dioxide reduction and acetate decarboxylation were the two primary methanogenic pathways, but the abundance of genes that encoded the key enzymes for the other two methanogenic pathways was quite low (Table S5). Acetate kinase (EC: 2.7.2.1), phosphate acetyltransferase (EC: 2.3.1.8) and acetate-CoA ligase (EC: 6.2.1.1) are the key enzymes for the conversion of acetic acid into acetyl-CoA. Acetyl-CoA is further converted into acetate decarboxylation pathway, and acetyl-CoA decarboxylase/synthase (ACDS) is the precursor of methane (5-methyl-THMPT) (Wang et al., 2018a). These enzymes are greatly improved after adding CaO_2 or magnetite. Especially in addition of CaO_2, the improvement is more obvious. This is consistent with the results of enzyme characteristics described in Section 3.3. Although the relative abundance of hydrogenotrophic methanogen community in the magnetite participated digesters (R1 and R3) is lower than that of R1 (Fig. 7), the abundance of genes encoding key enzymes detected in the carbon dioxide reduction pathway is higher. The most abundant enzyme is formylmethane furan dehydrogenase (EC1.2.7.12). The addition of CaO_2 alone did not significantly increase the number of genes encoding this enzyme, but the number in the magnetite participated digesters increased by more than 1.5 times higher than the control. The more actively expressed genes for CO_2-dependent methanogenesis show that the addition of magnetite strengthens the utilization of CO_2 by DIET through the magnetite.

3.6. Overall summary

In summary, the participation of a small amount of CaO_2 and magnetite in AD improved the digestive performance by improving the balance between fermentation bacterium and methanogenic archaea. The CaO_2 created a micro-aeration environment, which stimulated the growth of facultative bacteria. These bacteria are mainly involved in glycolysis, amino acid metabolism, TCA cycle, fatty acid metabolism, pyruvate metabolism and other pathways. At the same time, the enhancement of oxidative phosphorylation pathway of these bacteria provides more energy for transporting and transforming organic matter, and improves metabolic activity. The enhanced metabolic activity of these bacteria led to the improvement of the processes of hydrolysis and acidification, which induced the release of intermediates with a small molecular weight that can be utilized by methanogenic archaea. Under

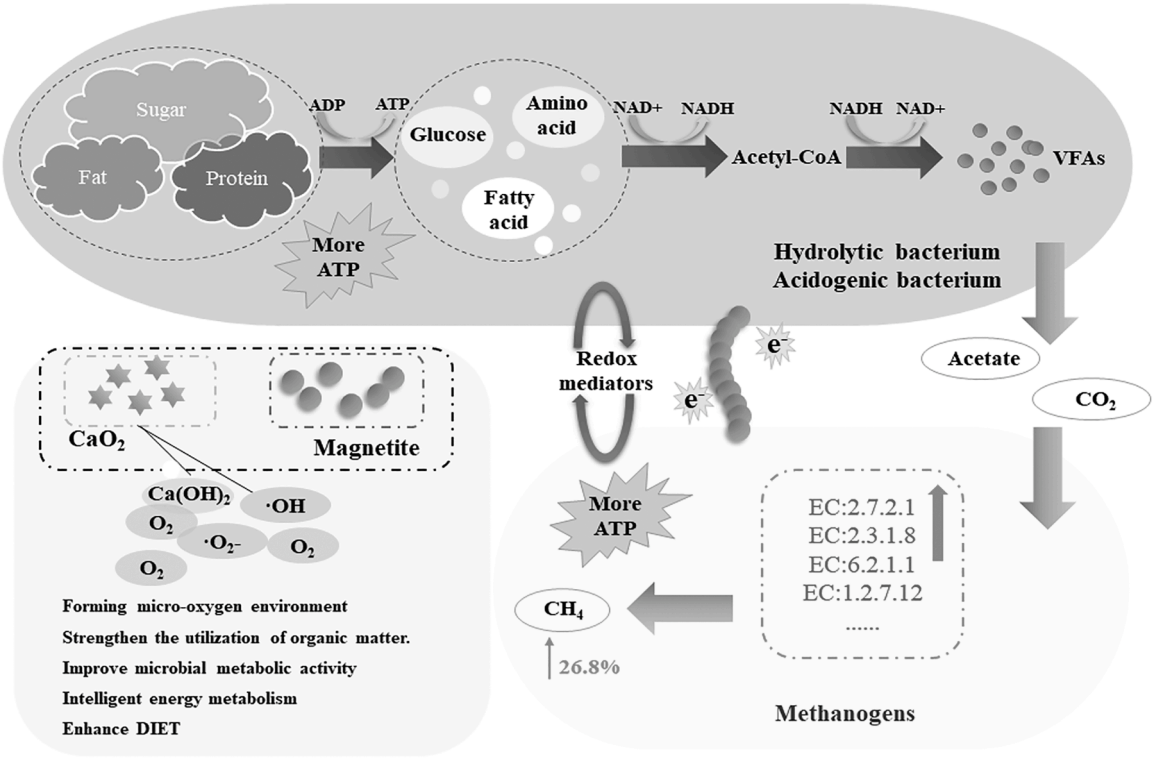


Fig. 10. Potential mechanism of CaO_2/magnetite addition in improving digestive performance in AD.

the co-effect of CaO_2 and magnetite, both bacteria and archaea showed increased metabolic activities, thus accelerating the conversion of organics to methane (Fig. 10). Altogether, the methane yield was significantly increased by the addition of CaO_2 and magnetite (26.8% higher than control).

Though the study was still in the laboratory research stage, the economic analysis results gave us confidence for potential engineering application (Table S7). CaO_2 is widely available. Its low price and limited dosage in AD further increase the possibility of engineering application. For magnetite, it can be magnetically separated from slurry and recycled. Compared with micro-aeration, microwave and heating treatments, direct introduction of additives into the digesters avoided the adjustment of the digestive equipment. In addition, the importance of the presence of trace electron acceptors in AD should be given more attention, as it appears critical in regulating the anaerobic microorganism community.

4. Conclusion

This study employed CaO_2 coupled with magnetite to regulate the digestive system by using FW as feedstock in semi-continuous AD. The results showed that: 1) The combination use of CaO_2 and magnetite improved the specific methane yield with the promotion ratio of 26.8%, as well as the conversion rate of organics to methane. 2) The activities of hydrolytic and acidogenic enzymes, F420 enzymes and electron transfer were significantly enhanced. The participation of CaO_2 and magnetite in AD enriched facultative bacteria and regulated the abundance distribution of methanogens. 3) CaO_2 addition increased the gene expression of hydrolysis and acidogenic pathway, accelerated TCA cycle and strengthened energy metabolism pathway, leading to the microbial adaptation and community structure optimization. 4) Magnetite acted as part of the electron carrier, reduce the synthesis energy of some enzymes related to oxidative phosphorylation, and strengthen the expression of the methane metabolism pathway. 5) Under the co-effect of magnetite and CaO_2, a better mutual nutrition relationship was built between bacteria and archaea.

Declaration of Competing Interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper and all authors have seen and approved the final version of the manuscript. They warrant that the article is the authors' original work, has not been published prior and is not under consideration for publication elsewhere.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2022.118801.

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